Department of Computer Science and Engineering

Course Code: CSE 433
Full Marks: 10
CT-01
Spring 2026

Course Title: Blockchain and Distributed Security
Duration: 30 minutes
Set: C (01)
Date: 29 August, 2026

*Answer all questions according to whether your **[ID]** is **[ODD]** or **[EVEN]***

Name:	
ID:	Section:

1. While distributed systems offer high fault tolerance, they introduce complex trade-offs. Identify two technical challenges unique to distributed systems that do not exist in traditional centralized architectures. **[2]**

2. *Considering Block N-1, N, and N+1 structure of a block series, explain how Block N+1 connects to Block N using block header references and state where the Merkle Root is stored within a block's overall layout.* **[3]**

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3. Suppose a blockchain development team is preparing two major network updates: [2]
Scenario 01: Changing consensus rules to introduce smart contracts and expand the block size limit.
Scenario 02: Introducing a more restrictive rule that reduces the block size limit and refines transaction formats.

If 30% of the network node operators refuse to upgrade their software for both updates, and considering backward compatibility, node behavior, and chain split risk:

- Explain what happens to the network structure for **Scenario 01** and **Scenario 02.**

[3]

4. In PoW Consensus mechanism:
   a. What specific privilege does a miner earn by being the first to solve the cryptographic puzzle?
   b. State what two independent checks a validator node performs when it receives a newly broadcast block.

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1. *Consider a private blockchain network running the Raft consensus protocol with 5 nodes (N1, N2, N3, N4, N5).* **[4]**

Assume, Node N3 is currently the elected Leader. Suddenly, N3 crashes. Explain the step-by-step transition process that Node N4 goes through to become the new leader, including the condition that triggers this change.

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2. What happens when the longest version of a blockchain starts spreading across the network? What specific name is given to a block that gets discarded when the network adopts a longer chain? [3]

3. What two main sections make up each block? List the specific items contained inside the Block Header. [3]

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